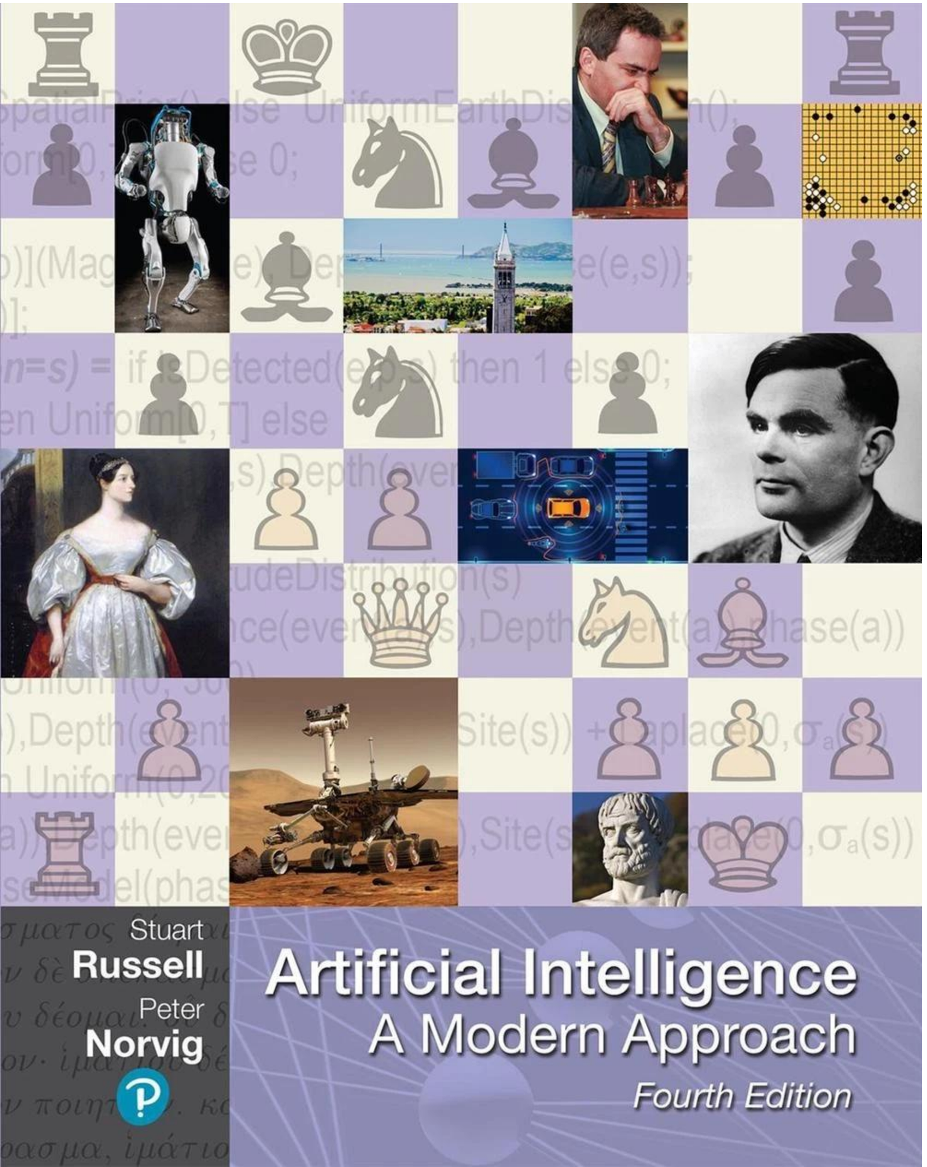
HOA SEN UNIVERSITY

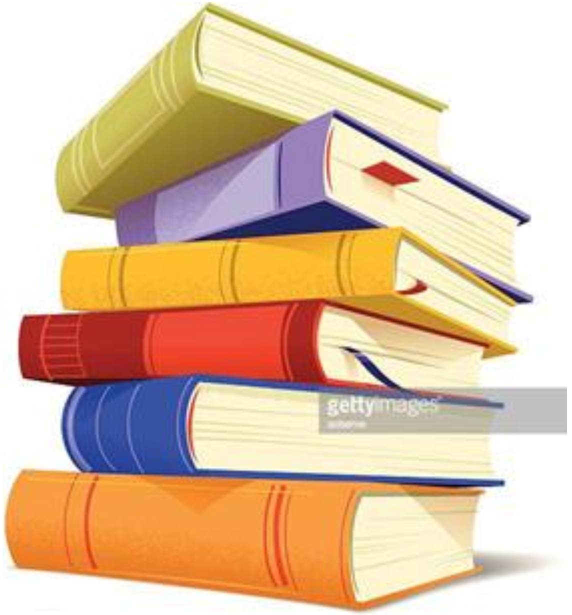
www.hoasen.edu.vn



Chapter 9

Deep Learning

Deep Learning



- Introduction to Deep Learning
- Perceptron & Multi-Layer Networks
- Activation Functions & Loss Functions
- Training Neural Networks

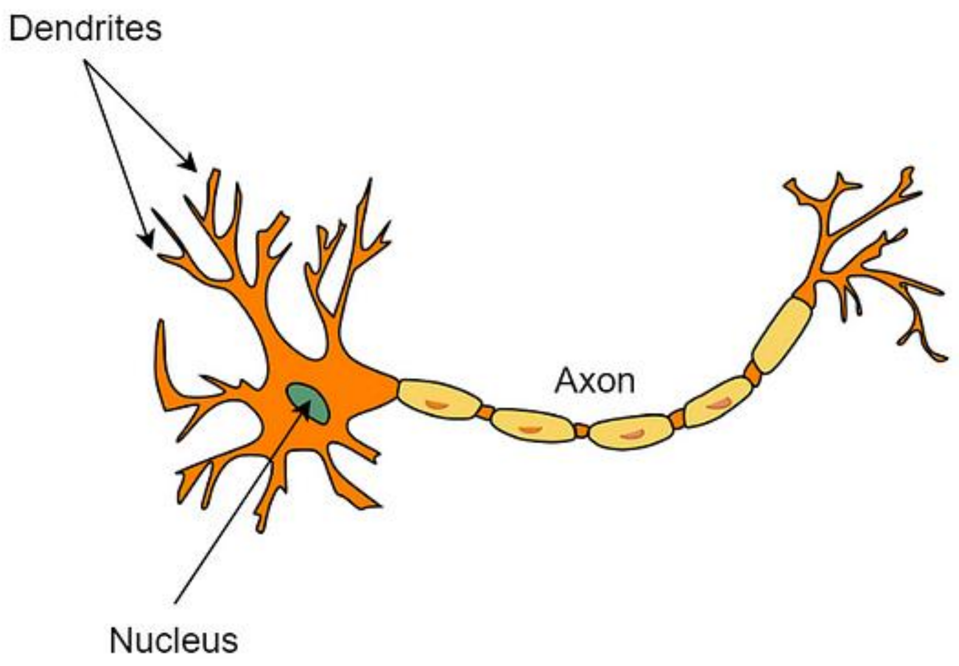
Introduction to Deep Learning

- Deep Learning is a subfield of Machine Learning that focuses on learning representations from data using deep neural networks.
- It has achieved breakthroughs in vision, speech recognition, and natural language processing.

Why Deep Learning?

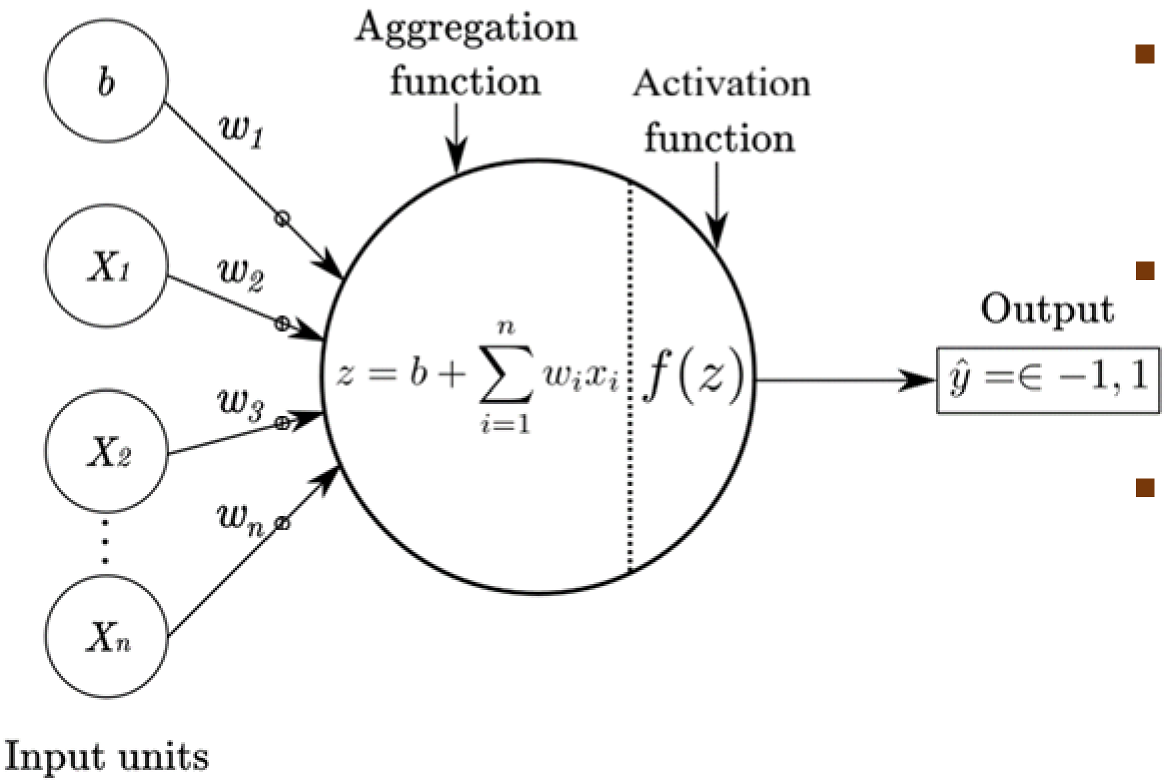
- Traditional Machine Learning relies heavily on manual feature engineering.
- Deep Learning automatically learns features from raw data.
- Large datasets and powerful hardware make Deep Learning practical.

Biological Inspiration



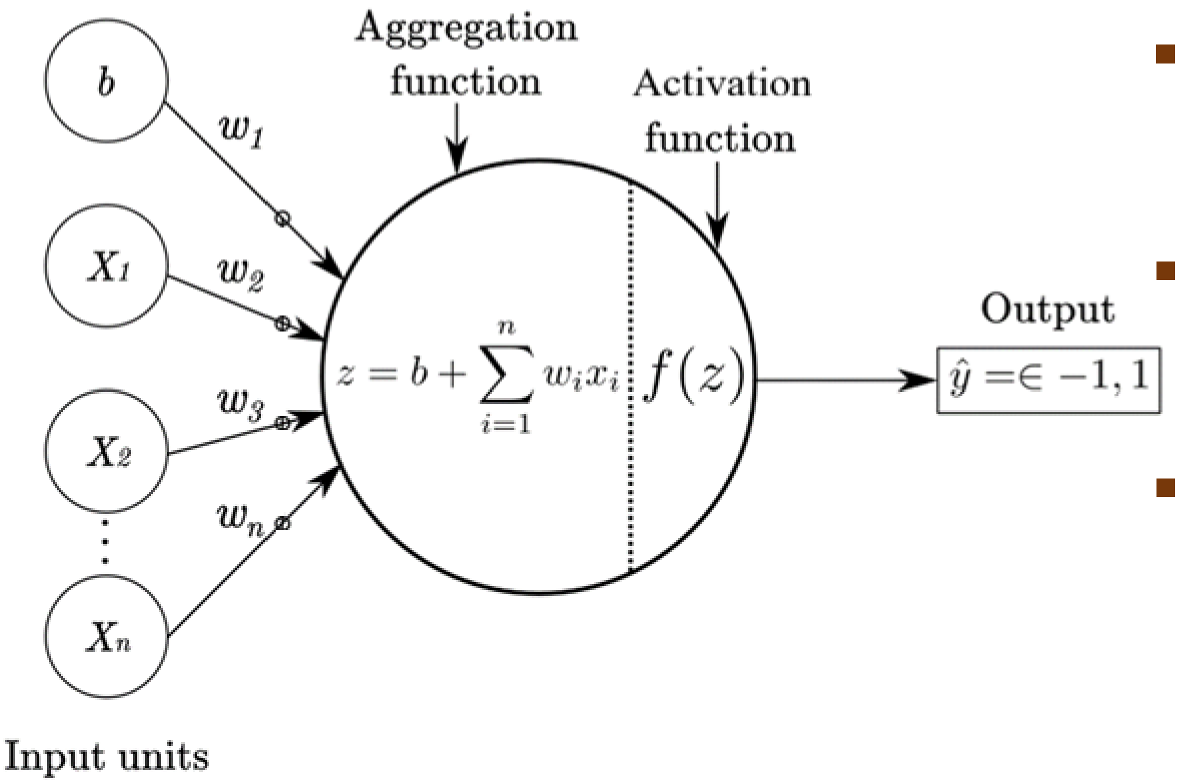
- Neural networks are inspired by the human brain.
- A biological neuron receives signals, processes them, and produces an output.
- Artificial neurons mimic this behavior mathematically.

Artificial Neuron Model



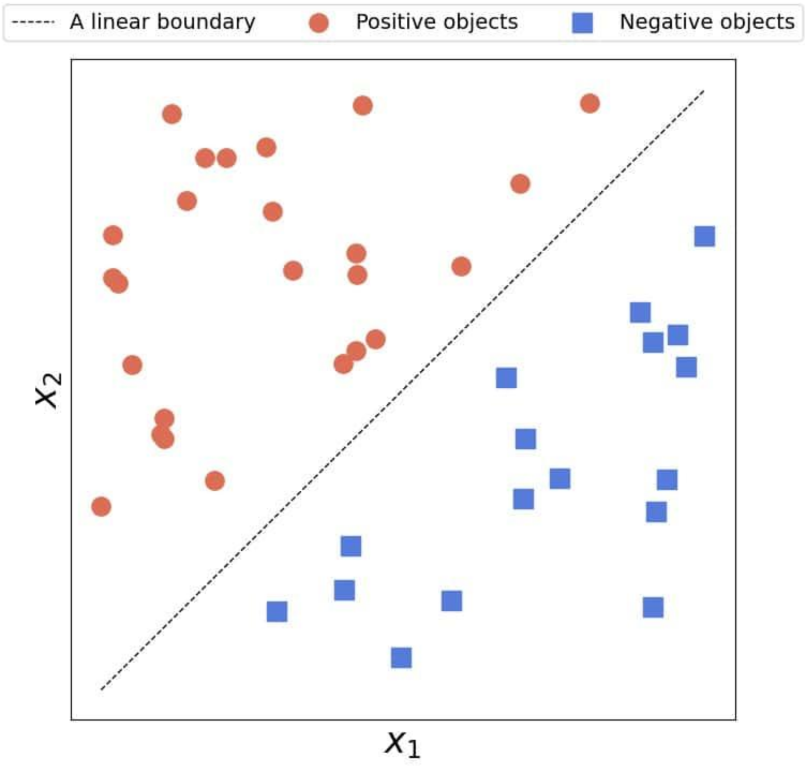
- An artificial neuron computes a weighted sum of inputs plus a bias.
- The result is passed through an activation function.
- This simple unit is the building block of deep networks.

Perceptron

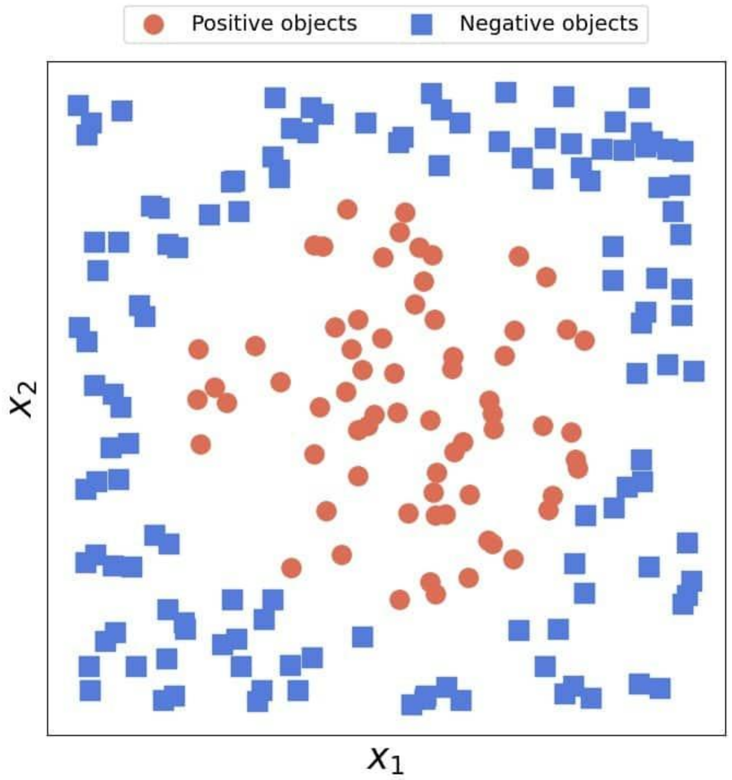


- The perceptron is the simplest neural network model.
- It can solve linearly separable problems.
- It fails on non-linear problems such as XOR.

Linearly & nonlinearly separable problems

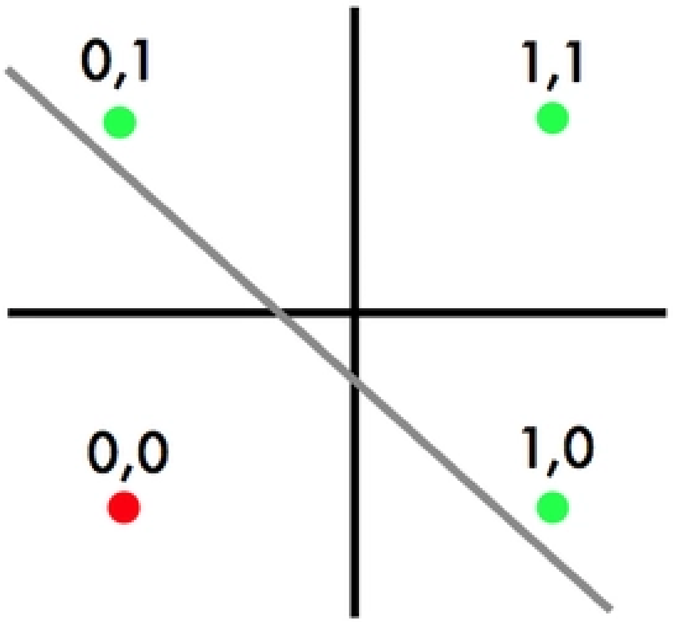


Linearly separable problem

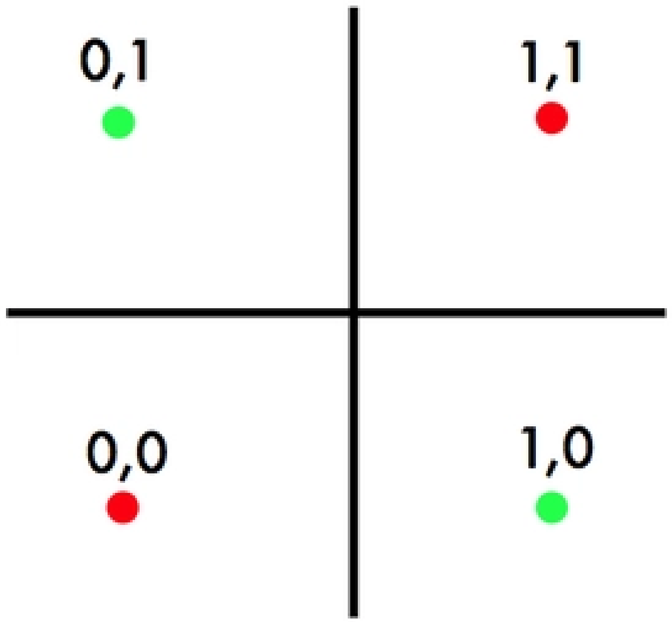


Nonlinearly separable problem

Linearly & nonlinearly separable problems

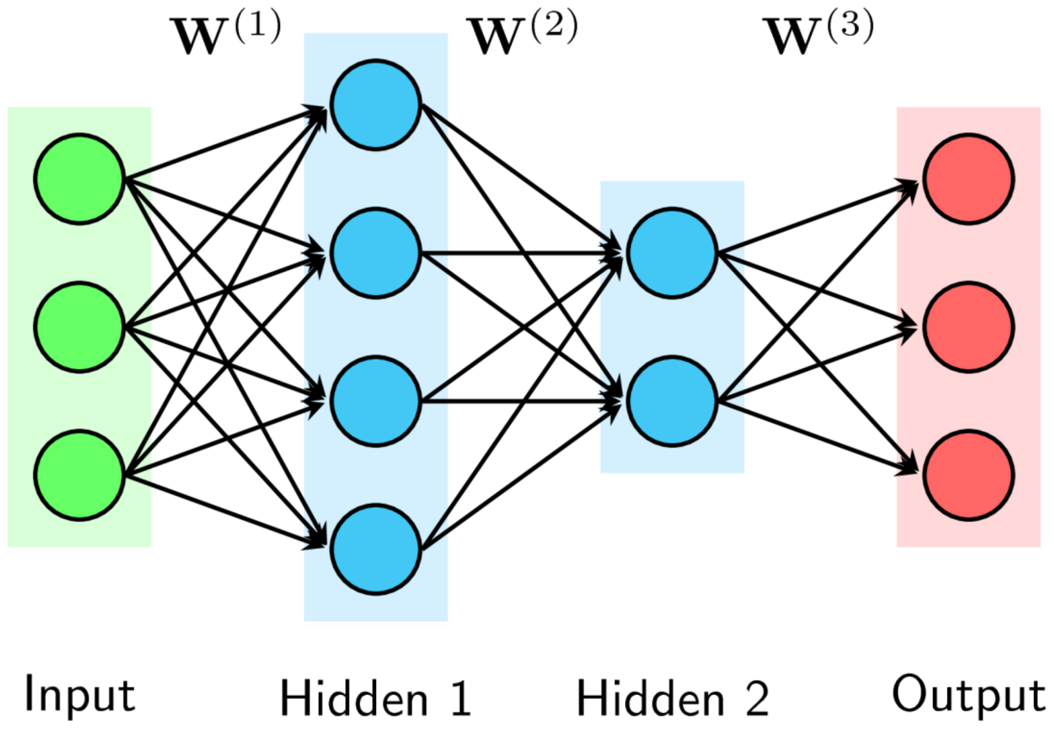


OR



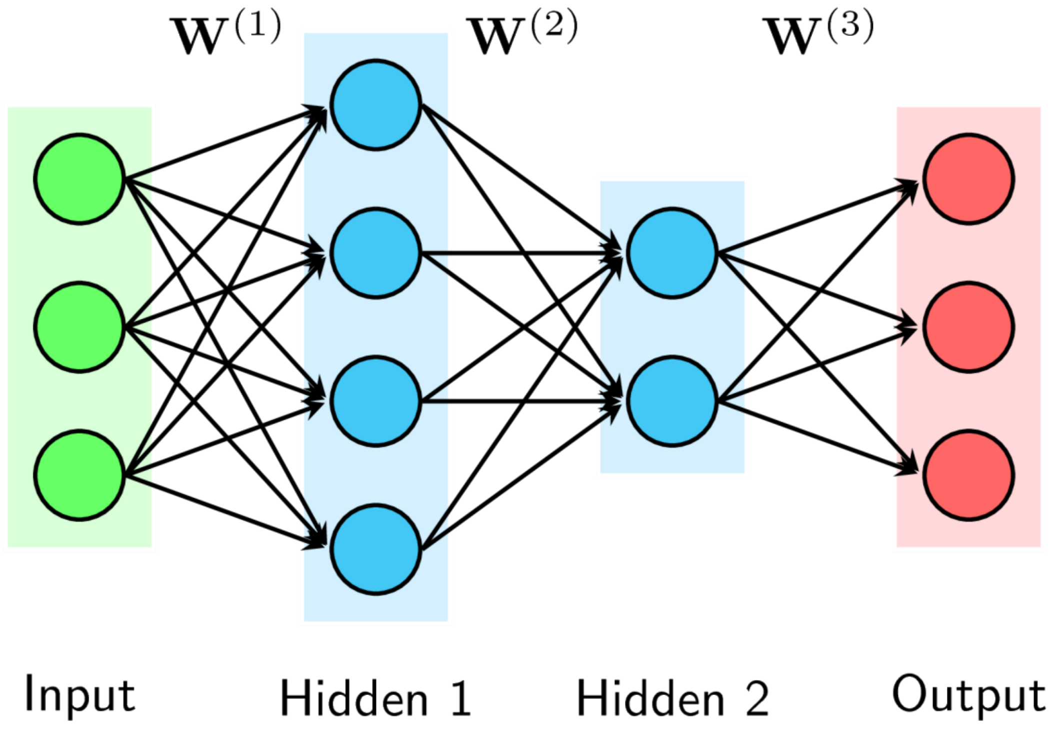
XOR

From Perceptron to Multi-Layer Networks



- Adding hidden layers allows modeling non-linear relationships.
- This leads to Multi-Layer Perceptrons.
- Depth increases representational power.

Forward Propagation



- Data flows from input layer to output layer.
- Each layer transforms the representation.
- The final output is the model prediction.

Activation Functions

- Activation functions introduce non-linearity, enabling MLPs to learn complex mappings.
- Common functions include Sigmoid, Tanh, ReLU, and Softmax.
- ReLU is widely used in deep networks.

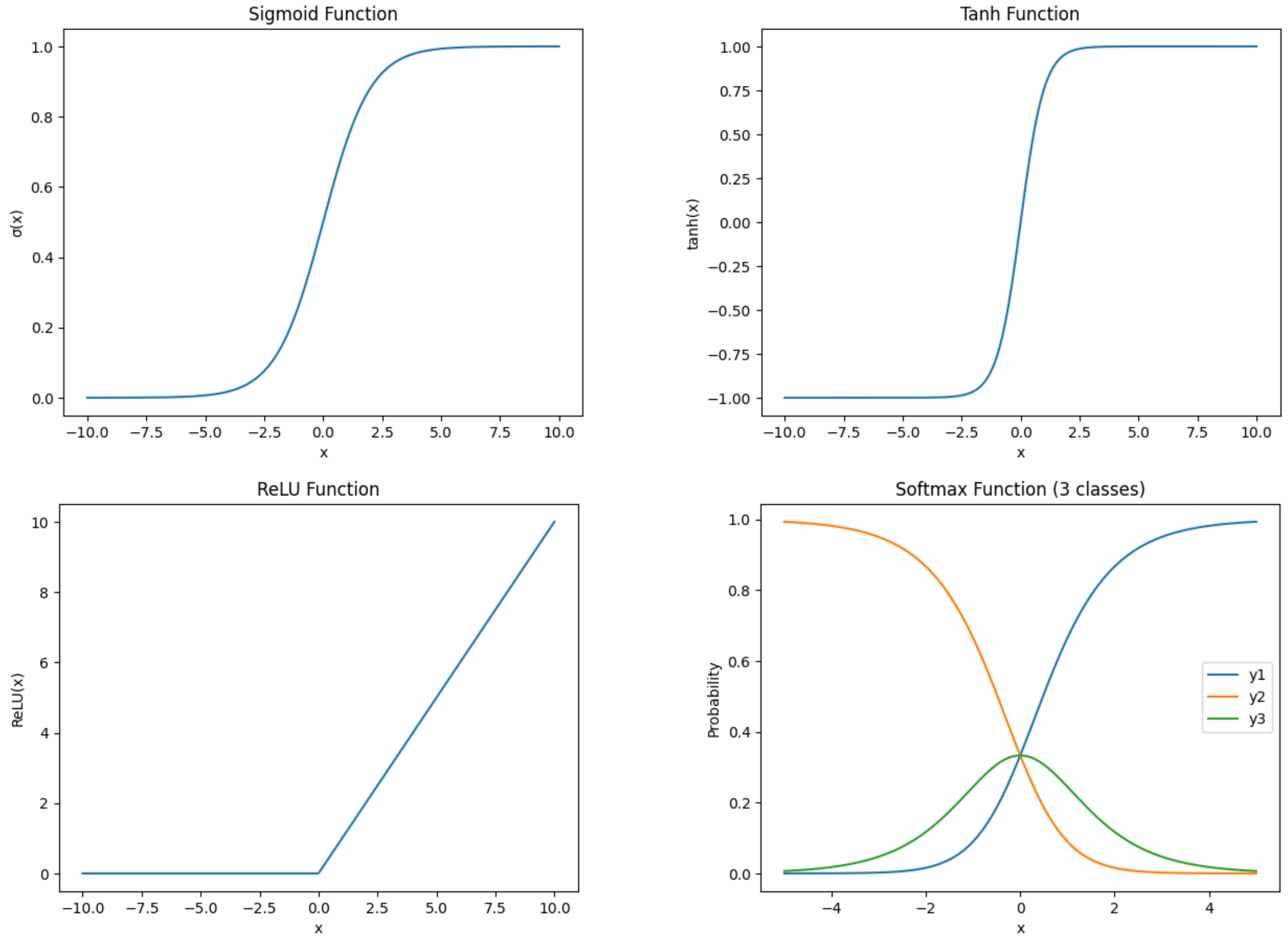
Common Activation Functions

Task	Activation Function	Expression	Characteristics
Binary Classification	Sigmoid	$f(x) = \frac{1}{1+e^{-x}}$	Outputs values in (0,1); interpretable as probability; suffers from vanishing gradients
Binary Classification (Hidden Layers, rarely)	Tanh	$f(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$	Outputs values in (−1,1); zero-centered; still affected by vanishing gradients
Hidden Layers	ReLU	$f(x) = \max(0, x)$	Simple and efficient; mitigates vanishing gradients; can suffer from dying ReLU problem
Multi-class Classification	Softmax	$f(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$	Outputs a probability distribution; class probabilities sum to 1; enables class competition

HOA SEN
UNIVERSITY

Common Activation Functions

Activation Function	Expression
Sigmoid	$f(x) = \dfrac{1}{1 + e^{-x}}$
Tanh	$f(x) = \dfrac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$
ReLU	$f(x) = \max(0, x)$
Softmax	$f(x_i) = \dfrac{e^{x_i}}{\sum_j e^{x_j}}$



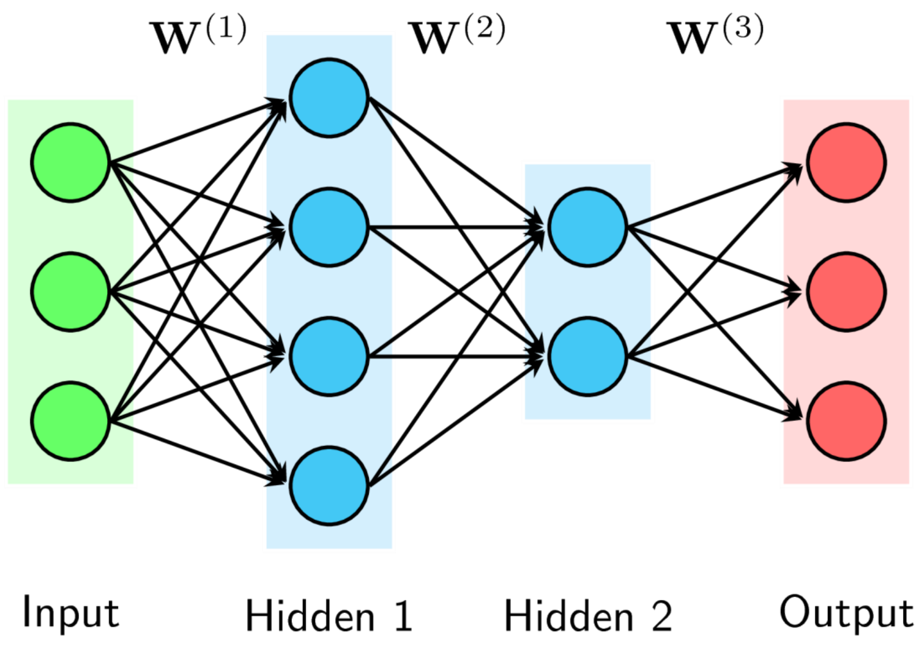
Loss Functions

- The loss function measures prediction error.
- Different tasks require different loss functions.
- Training aims to minimize the loss.

Task	Loss Function	Formula	Meaning
Regression	Mean Squared Error (MSE)	$L = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$	Penalizes large errors strongly by squaring the difference
Binary Classification	Binary Cross-Entropy	$L = -[y\log(p) + (1-y)\log(1-p)]$	Measures discrepancy between true labels and predicted probabilities
Multi-class Classification	Categorical Cross-Entropy	$L = -\sum_{k=1}^{K} y_k \log(p_k)$	Measures how well Softmax probabilities match the true class

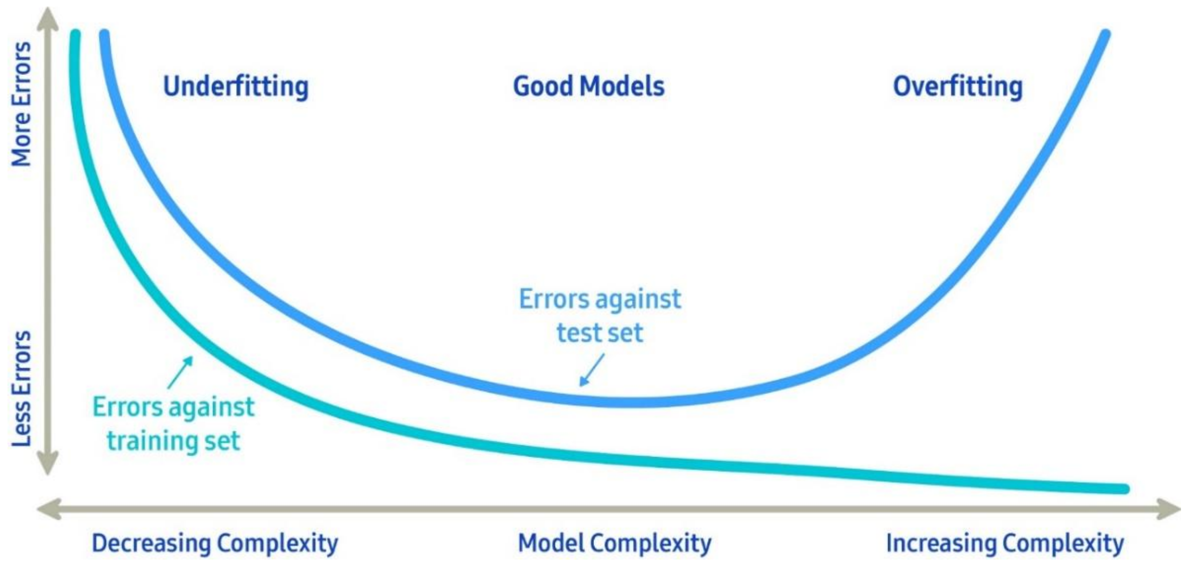
Training Neural Networks

- NNs are trained using the Backpropagation algorithm combined with Gradient Descent optimization:



1. **Forward Propagation:** Compute outputs layer by layer using current weights.
2. **Loss Computation:** Measure the difference between predicted and true outputs.
3. **Backpropagation:** Compute the gradient of the loss for each weight.
4. **Weight Update:** Adjust weights using optimizers such as **SGD**, **Adam**, or **RMSProp**.

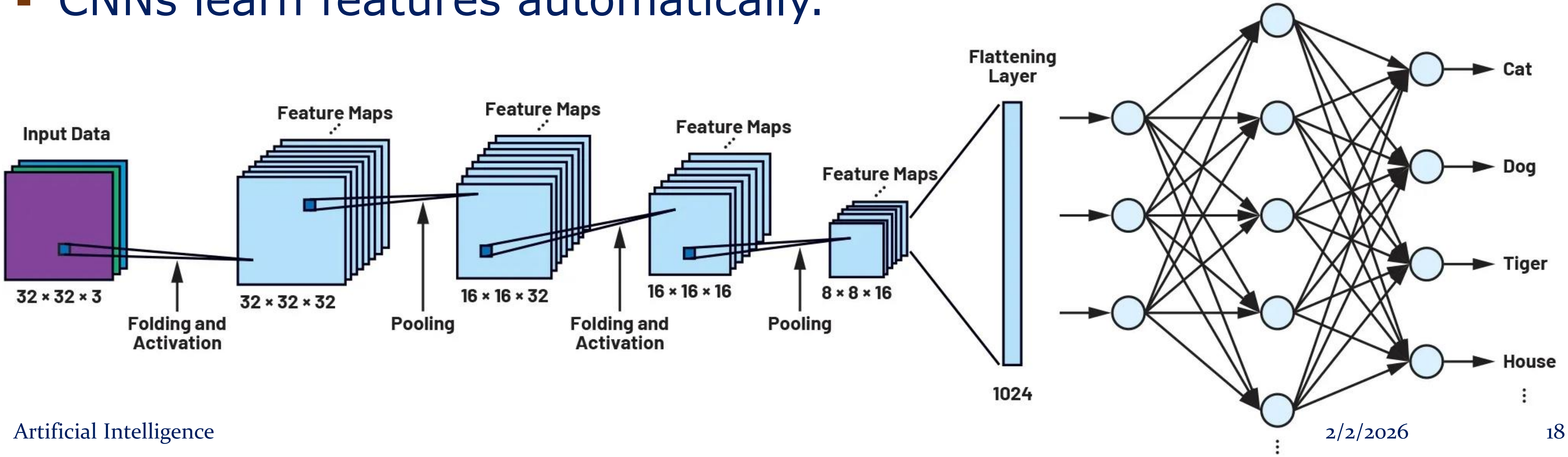
Overfitting and Underfitting



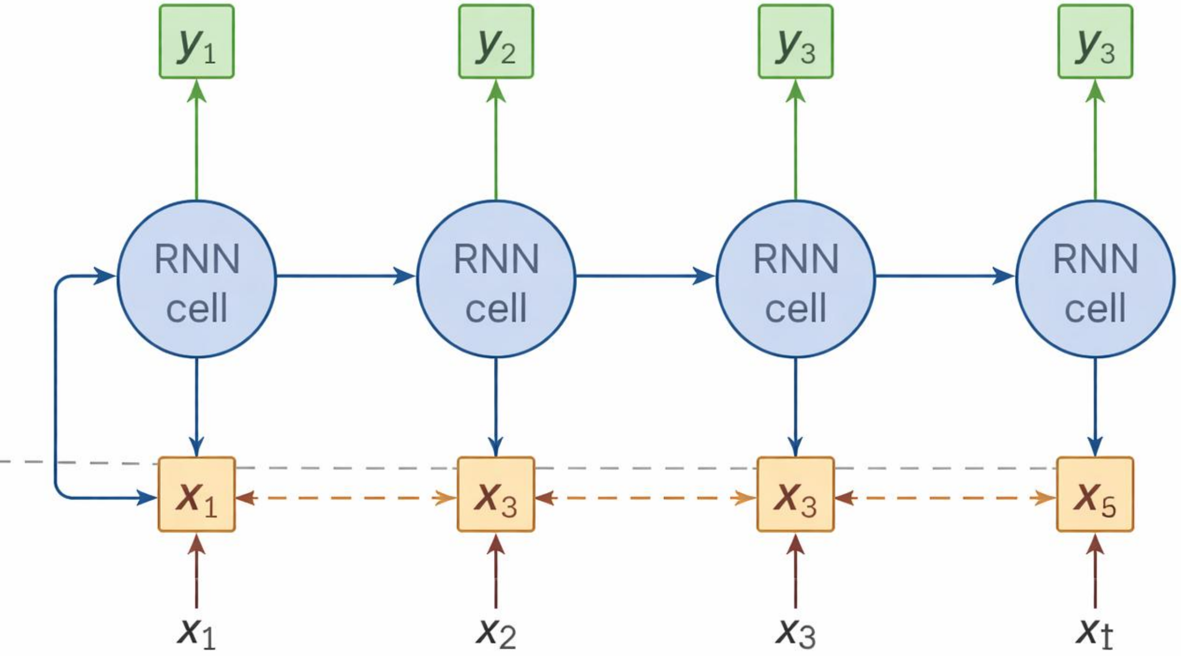
- Overfitting occurs when the model memorizes data.
- Underfitting occurs when the model is too simple.
- Regularization helps improve generalization.

Convolutional Neural Networks (CNNs)

- CNNs are designed for image data.
- They use convolution and pooling operations.
- CNNs learn features automatically.



Recurrent Neural Networks (RNNs)



- RNN is used to process sequential data, such as natural language processing and speech recognition.
- They maintain a hidden state over time.

Summary

- Deep Learning learns representations automatically.
- Neural networks are trained via backpropagation.
- Many architectures exist for different data types.


Q
A